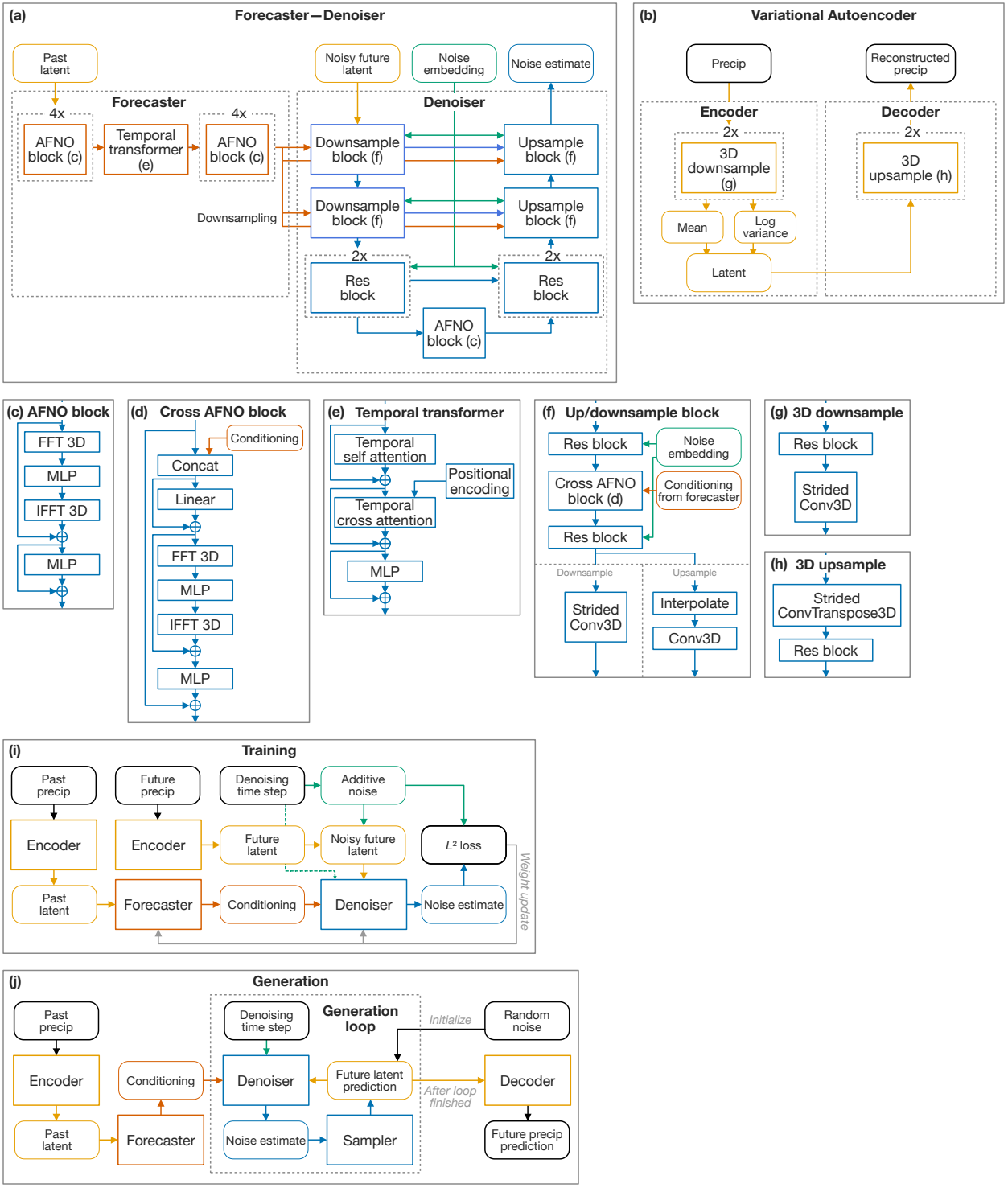
(a)
Forecaster—Denoiser
Past latent
Forecaster
4x
AFNO block (c)
Temporal transformer (e)
4x
AFNO block (c)
Noisy future latent
Noise embedding
Noise estimate
Denoiser
Downsample block (f)
Upsample block (f)
Downsampling
Downsample block (f)
Upsample block (f)
2x
Res block
2x
Res block
AFNO block (c)
(b)
Variational Autoencoder
Precip
Reconstructed precip
Encoder
2x
3D downsample (g)
Mean
Log variance
Latent
Decoder
2x
3D upsample (h)
(c) AFNO block
FFT 3D
MLP
IFFT 3D
MLP
(d) Cross AFNO block
Conditioning
Concat
Linear
FFT 3D
MLP
IFFT 3D
MLP
(e) Temporal transformer
Temporal self attention
Positional encoding
Temporal cross attention
MLP
(f) Up/downsample block
Res block
Noise embedding
Cross AFNO block (d)
Conditioning from forecaster
Res block
Downsample
Strided Conv3D
Upsample
Interpolate
Conv3D
(g) 3D downsample
Res block
Strided Conv3D
(h) 3D upsample
Strided ConvTranspose3D
Res block
(i)
Training
Past precip
Future precip
Denoising time step
Additive noise
Encoder
Encoder
Future latent
Noisy future latent
L^2 loss
Past latent
Forecaster
Conditioning
Denoiser
Noise estimate
Weight update
(j)
Generation
Past precip
Denoising time step
Generation loop
Initialize
Random noise
Encoder
Conditioning
Denoiser
Future latent prediction
After loop finished
Decoder
Past latent
Forecaster
Noise estimate
Sampler
Future precip prediction